

EVA

ESP32-BASED INTERACTIVE ROBOTIC HEAD

Sense → Decide → Move → Express

A compact embedded-systems project combining sensing, real-time firmware, motion control and visual feedback.

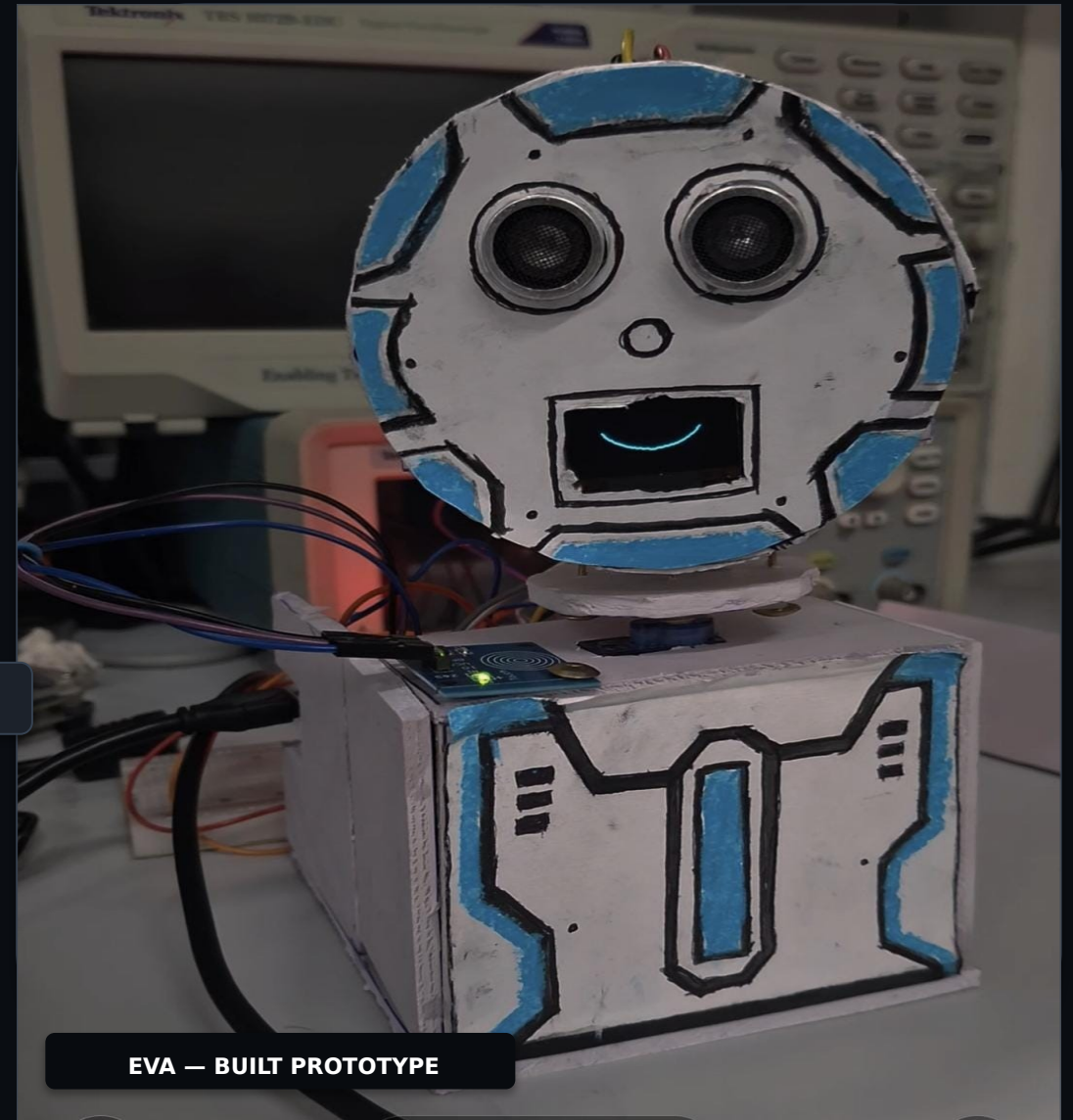
HC-SR04

OLED

SERVO

TOUCH

WI-FI



EVA — BUILT PROTOTYPE

IIT MADRAS • ELECTRONIC SYSTEMS

EVA • ESP32 Interactive Robotic Head

What EVA actually does

02

One system, four jobs

SENSE

HC-SR04
TTP223

PROCESS

Filtering
Detection + FSM

RESPOND

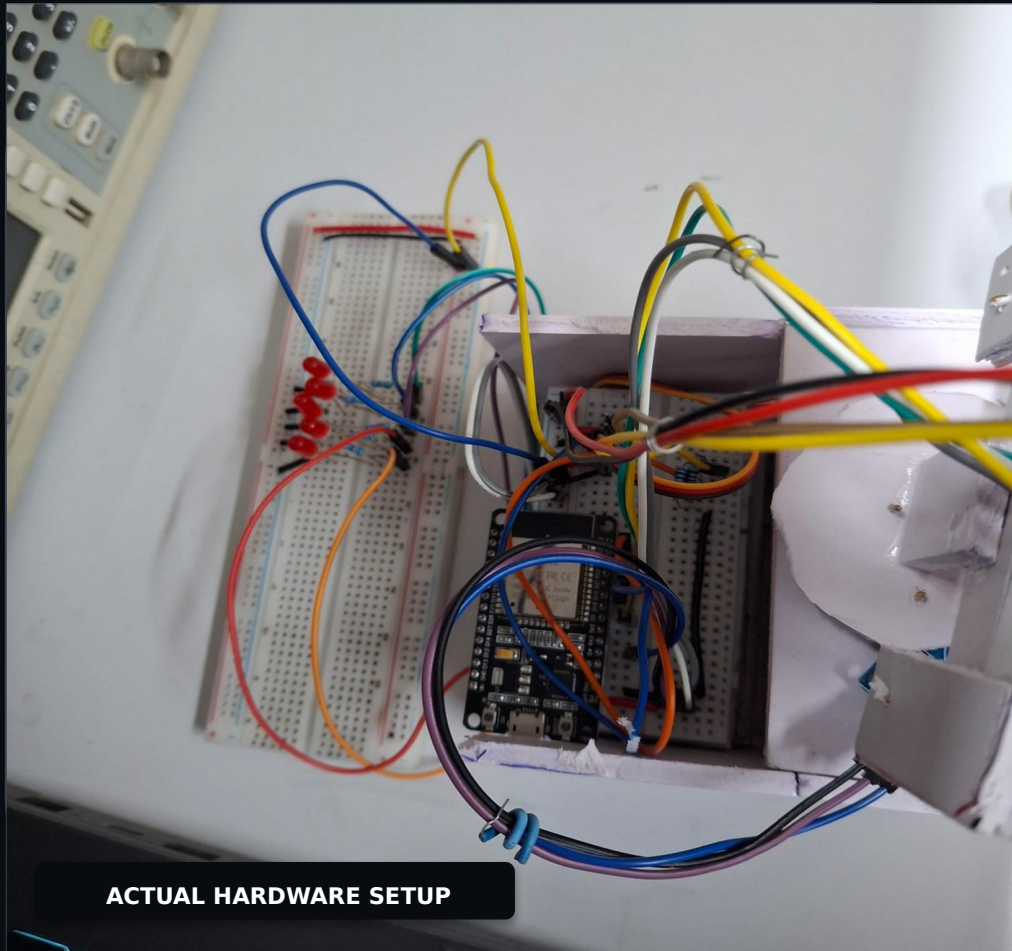
Servo motion
Expressions

VISUALIZE

OLED
Radar PC view

Hardware architecture

The physical system behind the firmware

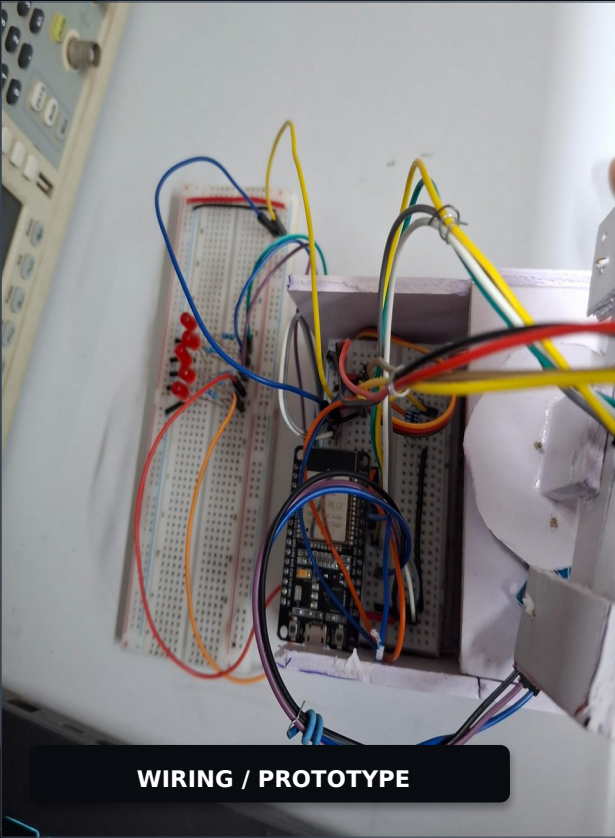


Actual build includes the controller, sensors, display, actuator and distance-indicator LEDs.

Interfaces & pin mapping

Keep the hardware wiring readable

Device	Interface	ESP32 pin
OLED SSD1306	I ² C	SDA 21 • SCL 22
HC-SR04	Digital	TRIG 5 • ECHO 23
TTP223	Digital	GPIO 4
Servo	PWM	GPIO 19
Distance LEDs	Digital	13, 12, 14, 26, 25



Design note
Servo is driven by the ESP32, while the physical build uses the external actuator supply.

Distance sensing: from echo to decision

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HC-SR04 measurement pipeline



Distance calculation

$$d = (t \times 0.0343) / 2$$

Example: 1000 µs → 17.15 cm

Why the filter matters

- Rejects isolated noisy readings
- Keeps detection stable
- Invalid readings are tracked separately
- Sensor fault after repeated bad readings

Reliable detection

EVA does not react to a single noisy measurement

NORMAL MODE

Person enters ≤ 60 cm

2 confirmations

Person leaves

3 confirmations + 900 ms

Result

EVENT_PERSON_ENTERED
EVENT_PERSON_LEFT

DISTANCE LED BAR

LED 1 ≤ 100 cm

LED 2 ≤ 80 cm

LED 3 ≤ 60 cm

LED 4 ≤ 40 cm

LED 5 ≤ 20 cm

FAULT HANDLING

- 3-200 cm = valid measurement range
- Invalid readings increment a bad-reading counter
- 8 consecutive bad readings → sensor fault
- Median filtering reduces jitter before detection

This is the difference between a demo and a more robust embedded implementation.

Behaviour architecture

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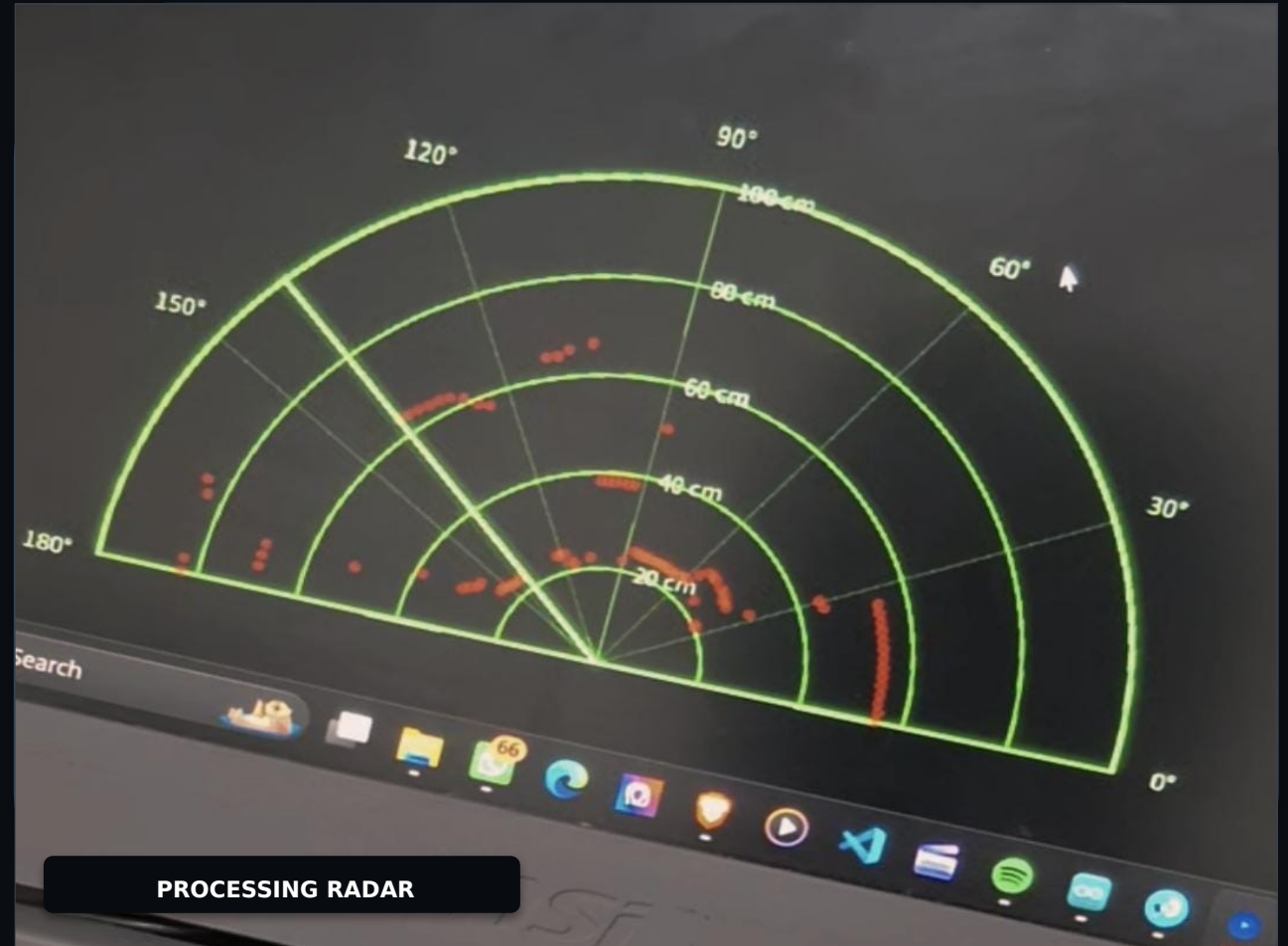
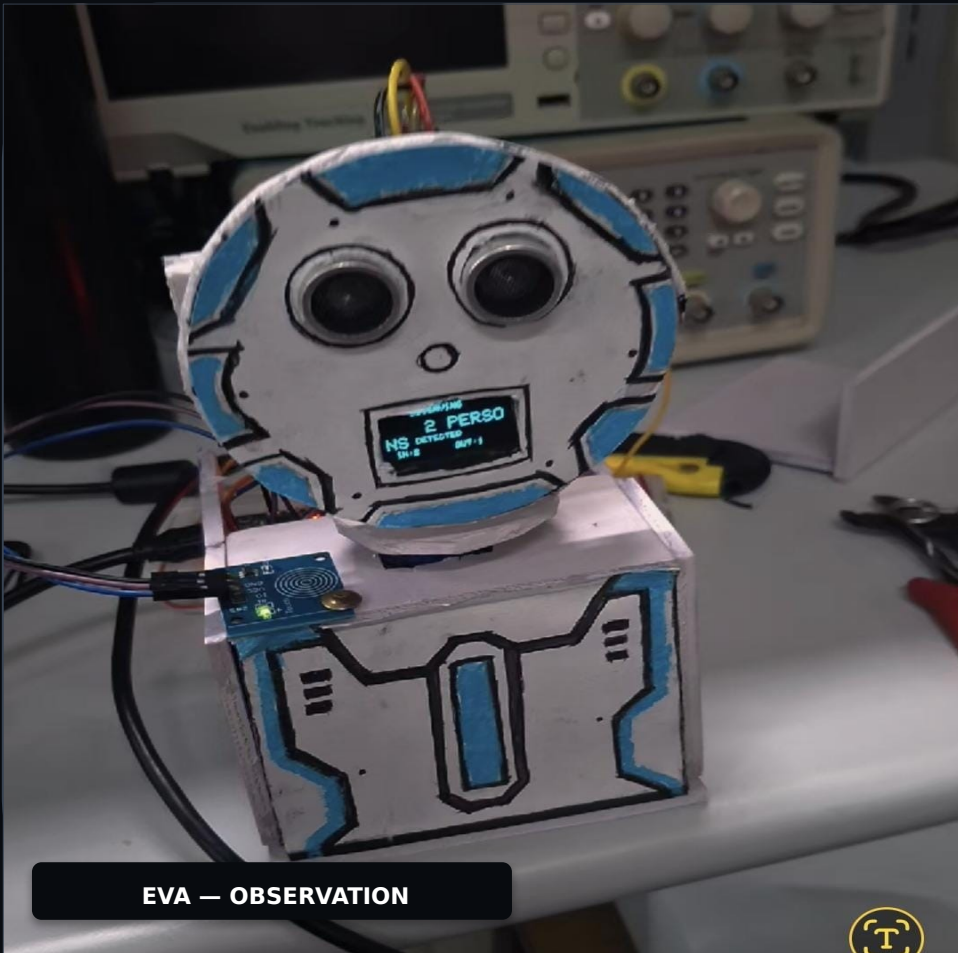
Touch → mode → events → expression



Observation mode + radar

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Scanning the environment and visualizing detections



Important: one HC-SR04 provides distance observations, not unique-person identity.

Firmware architecture

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Non-blocking scheduling keeps EVA responsive

MAIN LOOP

`schedulerUpdate()`

The loop stays small.
Tasks run when their period expires.

**No long blocking
delays in the main
EVA firmware.**

TASK PERIODS

Touch	20 ms
Servo	20 ms
Display	40 ms
Sensor	60 ms
Radar	60 ms
Telemetry	1 s

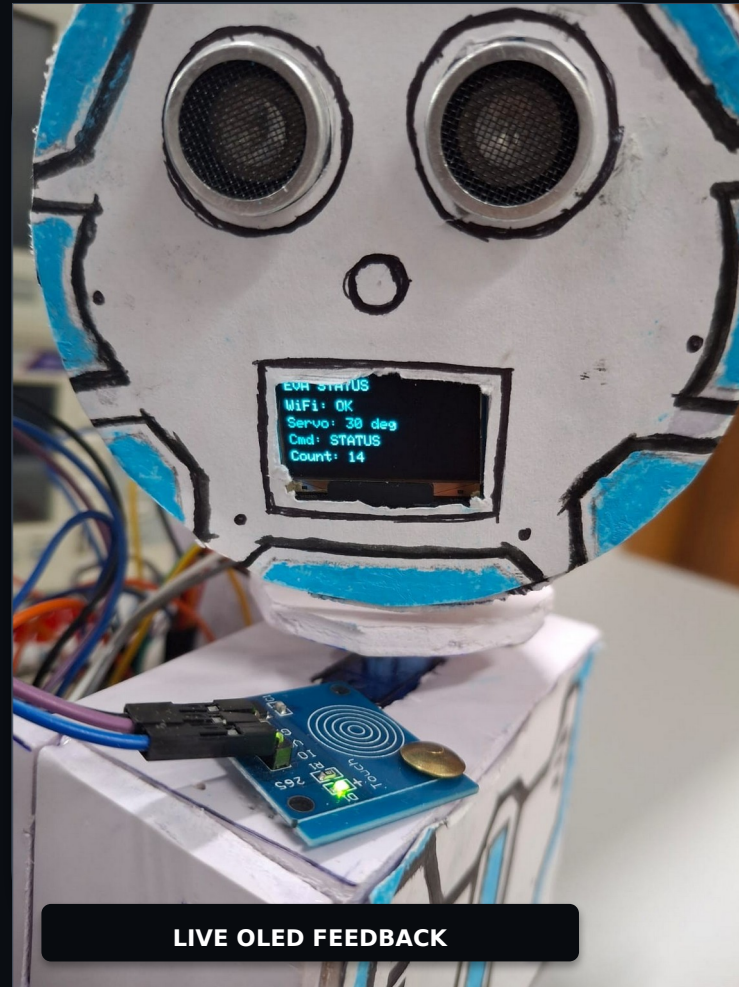
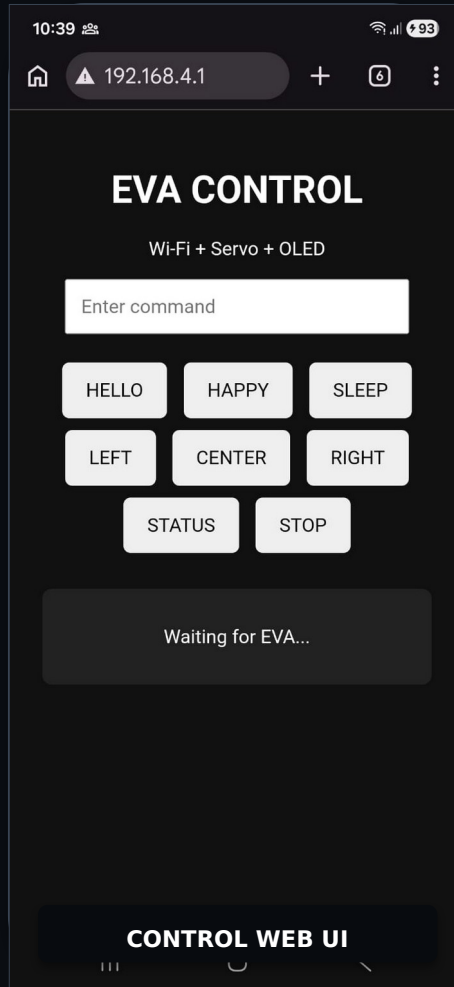
WHY IT MATTERS

- Touch stays responsive
- Servo motion continues
- OLED animation updates
- Ultrasonic echo is serviced
- Radar data can stream

**Core embedded-systems idea:
cooperative time-based scheduling.**

Phone Wi-Fi control

Command EVA straight from a browser, no app required



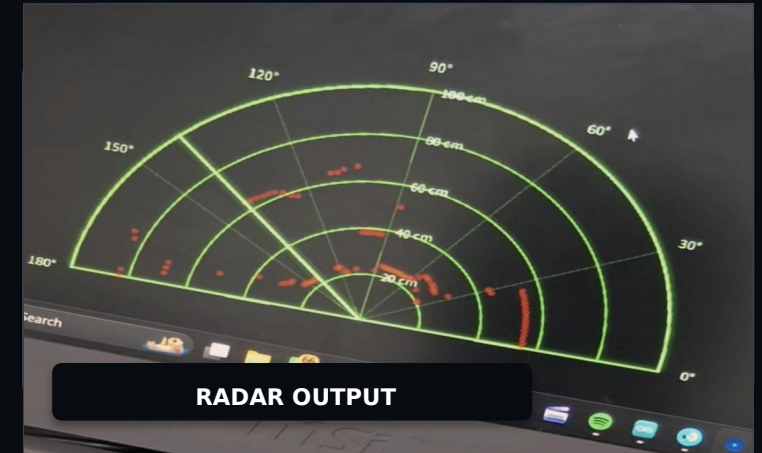
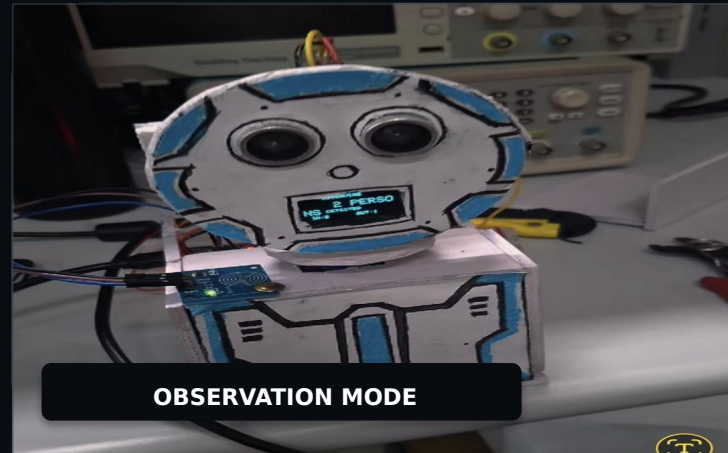
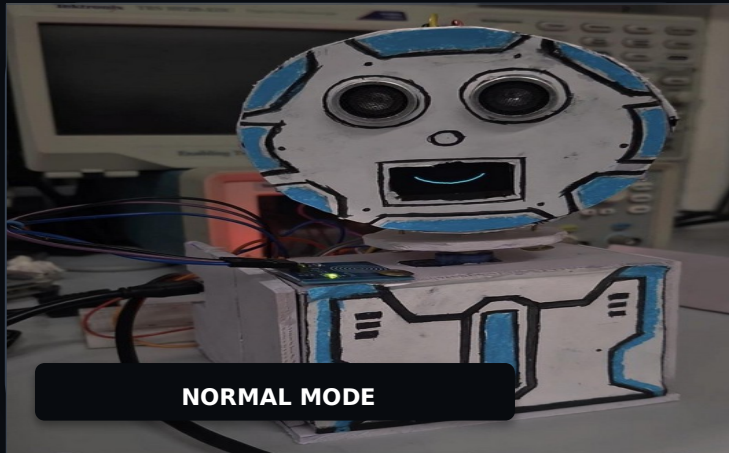
How it works

- ESP32 hosts its own Wi-Fi access point — no internet or app needed
- Phone connects to 192.168.4.1 and loads the control page
- Buttons send commands: HELLO, HAPPY, SLEEP, LEFT, CENTER, RIGHT, STATUS, STOP
- OLED echoes the live state back: Wi-Fi status, servo angle, last command, event count

Validation & prototypes

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What has been demonstrated so far



CURRENT DEMONSTRATION

- Physical EVA prototype with sensing, OLED, touch, servo and LEDs
- Normal + Observation modes with filtering and event-driven behaviour
- PC radar visualization from serial telemetry
- Phone Wi-Fi control via built-in ESP32 access point